| 19 | **Answer: A** |
|---|---|
| | change in potential energy = final potential energy – initial potential energy<br>$$=\left[\frac{qQ_1}{4\pi\varepsilon_0}\left(\frac{1}{x_1+x_2}\right)+\frac{q(-Q_2)}{4\pi\varepsilon_0}\left(\frac{1}{x_3}\right)\right]-\left[\frac{qQ_1}{4\pi\varepsilon_0}\left(\frac{1}{x_1}\right)+\frac{q(-Q_2)}{4\pi\varepsilon_0}\left(\frac{1}{x_2+x_3}\right)\right]$$<br>$$=\frac{qQ_1}{4\pi\varepsilon_0}\left(\frac{1}{x_1+x_2}-\frac{1}{x_1}\right)+\frac{qQ_2}{4\pi\varepsilon_0}\left(\frac{1}{x_2+x_3}-\frac{1}{x_3}\right)$$<br>work done by electric field = –(change in potential energy)<br>$$=\frac{qQ_1}{4\pi\varepsilon_0}\left(\frac{1}{x_1}-\frac{1}{x_1+x_2}\right)+\frac{qQ_2}{4\pi\varepsilon_0}\left(\frac{1}{x_3}-\frac{1}{x_2+x_3}\right)$$<br>**Note**: The change in potential energy is the work done by the external force. The electric force is equal and opposite to the external force (in defining the electric potential), hence there is an overall negative sign between the work done by the two forces. |

| 20 | **Answer: D** |
|---|---|